



Lab 04: Selection Statements

CSE 4108

Structured Programming I Lab

Lab Tasks

1. Counting Digits:

Students of International University of Tokyo (IUT) are famous for counting things. Anything you ask them, from flag stands, fields, departments, lakes, even leaves of trees, they can immediately tell you their count. The difficulty of such an answer depends on how many digits are in that count. The more the difficulty, the greater the reward. Alice, a freshman at IUT, wants to know the reward she's going to get from answering questions. She knows the answers but she wants an automated system to find out the difficulty of each answer. Being her friend, the responsibility has fallen on your hands to help her.

Write a program that calculates how many digits a given integer number contains.

Sample run:

```
Enter an integer number: 57
The number 57 has 2 digits
```

You may assume that the number has no more than four digits.

2. Time Converter:

Write a program that asks the user for a 24-hour time, then displays the time in 12-hour form, and if the input is not in the correct 24-hour format then it should print an error message:

Sample Run 1:

```
Enter a 24-hour time: 21:11
Equivalent 12-hour time: 9:11 PM
```

Sample Run 2:

```
Enter a 24-hour time: 24:60
ERROR: Input is not in proper 24-hour format
```

Hint: Be careful not to display 12 : 00 as 0 : 00.

3. **Broking Bad:**

When stocks are sold or purchased through a broker, the broker's commission often depends upon the value of the stocks traded.

Suppose that a broker charges the amounts shown in the following table:

Transaction size	Commission rate
Under \$2,500	\$30 + 1.7%
\$2,500–\$6,250	\$56 + 0.66%
\$6,250–\$20,000	\$76 + 0.34%
\$20,000–\$50,000	\$100 + 0.22%
\$50,000–\$500,000	\$155 + 0.11%
Over \$500,000	\$255 + 0.09%

The minimum charge is \$39

The *broker.c* program we had discussed in class asks the user to enter the amount of the trade, then displays the amount of the commission:

```
Enter value of trade: $30000
Commission: $166.00
```

Now, modify the *broker.c* program by making both of the following changes:

(a) Ask the user to enter the number of shares and the price per share, instead of the value of the trade. The total value of the trade is the product of number of shares and price per share.

(b) Add statements that compute the commission charged by a rival broker (\$33 plus 3 cents (\$1 = 100 cents) per share for fewer than 2000 shares; \$33 plus 2 cents per share for 2000 shares or more).

Finally, display the rival's commission as well as the commission charged by the original broker.

Sample Run:

```
Enter number of shares: 2000
Enter price per share: $15
Original Broker Commission: $166.00
Rival Broker Commission: $73
```

4. Beaufort Scale:

Wind force is estimated by Beaufort Scale. Here's a simplified version of the scale:

<i>Speed (knots)</i>	<i>Description</i>
Less than 1	Calm
1-3	Light air
4-27	Breeze
28-47	Gate
48-63	Storm
Above 63	Hurricane

Write a program that asks the user to enter a wind speed (in knots), then displays the corresponding description.

5. Condition Calamity:

Write a program that finds the largest and smallest of four integers entered by the user:

```
Enter four integers: 21 43 10 35
Largest: 43
Smallest: 10
```

Do not use any logical (&&, ||, !) operators.

6. **Early Date:**

Write a program that prompts the user to enter two dates and then indicates which date comes earlier on the calendar.

Sample run:

Enter first date (mm/dd/yy): 12/21/19

Enter second date (mm/dd/yy): 2/16/20

12/21/19 is earlier than 2/16/20